

國立暨南國際大學資訊工程學系學士班必選修科目一覽表

Undergraduate Program Course Catalog of Department of Computer Science and Information Engineering, National Chi Nan University

本表(113)學年度入學新生適用 Effective for students admitted in Fall, 2024

最低畢業學分：128 學分，全校共同課程 16 學分，通識領域課程 15 學分

系必修 49 學分，專業選修 42 學分，自由學分 6 學分。 Undergraduate requirements: a minimum of 128 required credits, including 16 credits in the School-Wide Curriculum, 15 credits of General Education Courses, 49 credits of Departmental Required Courses, 42 credits of Elective Courses, and 6 credits of School Elective Courses.

壹、全校共同課程 1.School-Wide Curriculum			
一、共同課程（必修16學分） A. School-Wide Required Courses (16 credits)			
課程代號 Course Code	課程名稱 Course Name	學分數 Credits	備註 Notes
	跨域專業學術英文(一)：文本解析與書寫 Cross-disciplinary ESAP: Textual Interpretation and Composition	2	自110學年度起，英文能力分班改為依照本校自辦英檢(OOPT)作依據，分為三級上課：前10%的同學將分配至「跨域專業學術英文(一)：文本解析與書寫(進階)/跨域專業學術英文(二)：思辨與表達(進階)/跨域專業學術英文(三)：跨文化溝通與表達(進階)」上課；中間80%的同學將分配至原班英文上課；後10%的同學將分配至「跨域專業學術英文(一)：文本解析與書寫(基礎)/跨域專業學術英文(二)：思辨與表達(基礎)/跨域專業學術英文(三)：跨文化溝通與表達(基礎)」上課。 Since the first semester of 110 Academic Year (September 2021), the competence-based class grouping policy of Freshmen English (Course Title: Cross-disciplinary ESAP: Textual Interpretation and Composition) will be based on registered student's OOPT scores; Students will be accordingly divided into three different levels: advanced (top 10%), general (middle 80%) and basic (bottom 10%).
	跨域專業學術英文(二)：思辨與表達 Cross-disciplinary ESAP: Critical Thinking and Expression	2	
	跨域專業學術英文(三)：跨文化溝通與表達 Cross-disciplinary ESAP: Cross-Cultural Communicative Skills	2	
	中文思辨與表達一：閱讀書寫 Critical Thinking and Expression In Chinese (I) : Reading And Writing	2	僑外生修習「僑外生華語文」可抵通識人文領域課程，不得以「僑外生華語文」抵免中文思辨與表達課程。(Overseas students who take Chinese for Overseas Students can be credited in field of Humanities, but cannot be certified as Critical Thinking And Expression In Chinese.)
	中文思辨與表達二：多元應用 Critical Thinking and Expression In Chinese (II) : Multiple Applications	2	
	社會服務學習 Service Learning	1	依所屬學系分班上課，無特殊原因不得跨班選課。(Students are divided into groups by their own department, without any proper reasons they cannot change the decision.)
	資訊科技 Information Technology	1	每學年於上學期開課，不分學系上課。(Freshmen take courses in the first semester of each academic year, regardless of department.)
	大一體育(上)Physical Education (1)	1	上學期依所屬學系分班上課，無特殊原因不得跨班選課，惟未達40人之學系、院學士班班級，則採合併班開課；下學期則依運動項目分班，自由選課。(In the first semester of freshman, students are arranged to take Physical Education (1) by their original department.

	大一體育(下)Physical Education (2)	1	Without any proper reasons, change will be prohibited. If a class of the department or the interdisciplinary program that is less than 40 students, the combined class will be adopted. In the second semester of freshman, students are able to take Physical Education (2) by their interesting.)
	特色運動 (Featured Sports)	2	大二開始，每學期限修一項，畢業前須完成至少2項特色運動 (Students are limited take one course each semester since Sophomore, At least 2 featured sports courses are required before graduation.)
通識講座 General Education lecture program		0	畢業前至少應出席6場。(Students are required to attend at least 6 lectures before graduation.)

二、共同選修課程（各系依共同選修加列課程） B. School-Wide Elective Courses

	第二外語 (Second Foreign Language)		含日文、韓文、法文、德文及西班牙文等。(Including Japanese, Korean, French, German and Spanish, etc.)
	東南亞語言(Southeast Asia Languages)		含泰語、越南語、緬甸語、印尼語及馬來語，可抵特色通識領域。(Including Thai, Vietnamese, Burmese, Indonesian and Malay. These courses can be credited course credits in Topical General Education)
	進階語文(Advanced English)		
	其他選修(Other Elective Course)		

貳、通識領域課程（至少15學分） 2.General Education Courses (15 credits)

領域 Field (須修習資訊科技與自主學習課程) (Information Technology and Self-directed Learning courses are required)		學分數 Credits	備 註 Notes
人文領域 Humanities	文學與藝術(Literature and Art)	3	至少修習3學分。(Students should attend a total of at least 3 credits.)
	歷史哲學與文化(History, Philosophy and Culture)		
	AI文創協作(AI Collaboration in Cultural Creativity)		
社會領域 Society	法政與教育(Law, Politics and Education)	3	至少修習3學分。(Students should attend a total of at least 3 credits.)
	社經與管理(Society, Economy and Management)		
	資料科學分析(Data Science Analysis)		
自然領域 Nature	工程與科技(Engineering and Technology)	3	至少修習3學分。(Students should attend a total of at least 3 credits.)
	生命與科學(Life and Science)		
	程式設計(Programming)		
特色通識領域 Characteristic General Education	國際連結(International Connection)	3	至少修習3學分。(Students should attend a total of at least 3 credits.)
	淨零永續(Net-Zero Sustainability)		
	社會創新(Social Innovation)		

通識各主領域課程(人文、社會、自然及特色通識)得單一領域跨域認抵，至多4學分 (需經通識教育中心認定之)
 General education courses from various fields (humanities, social, natural, and topical general education) can be cross-credited within a single field, up to a maximum of 4 credits (subject to approval by the General Education Center).
 領域課程15學分中，學生須至少修習二門資訊科技課程，且須包括自主學習類型課程1學分。
 Students are required to take at least two Information Technology courses and get 1 credit of Self-directed Learning courses out of the 15 credits of General Education Courses.

參、系必修（至少49學分） 3.Departmental Required Courses (49 credits)					
課程代號 Course Code	課程名稱(中文及英文) Course Name (Chinese and English)	學分數 Credits	修別 Course type (required/elective)	開課年級 Course availability	備註 Notes
210036	計算機概論 Introduction to Computer Science	3	必 (R)	一上 Freshman 1st Semester	
210003	微積分（上） Calculus(1)	3	必 (R)	一上 Freshman 1st Semester	
210013	離散數學 Discrete Mathematics	3	必 (R)	一上 Freshman 1st Semester	
210008	微積分（下） Calculus(2)	3	必 (R)	一下 Freshman 2nd Semester	
210099	程式設計 Programming	3	必 (R)	一下 Freshman 2nd Semester	
210151	邏輯設計與實驗（一） Logic Design and Laboratory Experiments (1)	3	必 (R)	一下 Freshman 2nd Semester	
210111	機率 Probability	3	必 (R)	一下 Freshman 2nd Semester	
210016	資料結構與演算法（一） Data Structures and Algorithms (I)	3	必 (R)	二上 Sophomore 1st Semester	
210152	邏輯設計與實驗（二） Logic Design and Laboratory Experiments (2)	3	必 (R)	二上 Sophomore 1st Semester	
210022	資料結構與演算法（二） Data Structures and Algorithms (II)	3	必 (R)	二下 Sophomore 2nd Semester	

210026	線性代數 Linear Algebra	3	必 (R)	二下 Sophomore 2nd Semester	
210025	計算機組織與結構 Computer Organization and Architecture	3	必 (R)	三上 Junior 1st Semester	
210093	系統程式 System Programming	3	必 (R)	三上 Junior 1st Semester	
210121	專題（一） Project Study (I)	2	必 (R)	三上 Junior 1st Semester	
210122	專題（二） Project Study (II)	2	必 (R)	三下 Junior 2nd Semester	
210102	作業系統 Operating Systems	3	必 (R)	三下 Junior 2nd Semester	
210160	微電腦實驗 Microcomputer Systems Lab	3	必 (R)	三下 Junior 2nd Semester	

肆、專業選修（至少42學分） 4.Elective Courses (42 credits)

210009	基本電學 Basic Electricity	3	選 (E)	一 Freshman	
210012	資訊系統與網路導論(二) Introduction to Computer System and Network II	3	選 (E)	一 Freshman	
210015	數位電子學 Digital Electronics	3	選 (E)	一 Freshman	
210021	組合數學 Combinatorial Mathematics	3	選 (E)	一 Freshman	
210023	數位電路實驗 Digital Circuits Lab	1	選 (E)	二 Sophomore	
210123	自動機與形式語言 Automata and Formal Language	3	選 (E)	三 Junior	
210124	數位系統設計 Digital System Design	3	選 (E)	一 Freshman	
210125	數值方法 Numerical Methods	3	選 (E)	一 Freshman	
210126	電腦視覺與人機介面 Computer Vision and Man-machine Interface	3	選 (E)	一 Freshman	
210130	具體數學 Concrete Mathematics	3	選 (E)	一 Freshman	
210131	UNIX程式設計 UNIX Programming	3	選 (E)	一 Freshman	
210132	程式語言 Programming Languages	3	選 (E)	一 Freshman	

210133	隨機程序 Introduction to Fundamental Stochastic Processes	3	選 (E)	二 Sophomore	
210134	LISP程式設計 LISP Programming	3	選 (E)	一 Freshman	
210135	UNIX使用入門 Introduction to Unix Environment	3	選 (E)	一 Freshman	
210136	電腦網路通信協定 Computer Network Protocols	3	選 (E)	一 Freshman	
210140	智慧型行動裝置軟體設計 Intelligent Mobile Device Programming	3	選 (E)	一 Freshman	
210141	3D建模與互動多媒體應用 3D Modeling and Interactive Multimedia Applications	3	選 (E)	一 Freshman	
210142	數值分析 Numerical Analysis	3	選 (E)	一 Freshman	
210143	Web服務和應用 Web Services and Applications	3	選 (E)	一 Freshman	
210146	基礎程式設計 Computer Fundamentals &	1	選 (E)	一 Freshman	
210147	基礎程式設計 (二) C++ Programming Fundamentals	1	選 (E)	一 Freshman	
210148	專業實習 Specialized Internship	3	選 (E)	四 Senior	
210149	企業實習 Enterprise Internship	6	選 (E)	四 Senior	
210009	電子電路 Electrical Circuits	3	選 (E)	一 Freshman	
210086	工程數學 Engineering Mathematics	3	選 (E)	二 Sophomore	
210115	邏輯設計 Logic Design	3	選 (E)	一 Freshman	
210145	計算機網路 Computer Networks	3	選 (E)	一 Freshman	
210029	微算機系統 Microcomputer Systems	3	選 (E)	一 Freshman	
210153	MFC視窗程式設計 Windows MFC Programming	3	選 (E)	一 Freshman	需先修程式設計 prerequisite: Programming
210154	個人電腦概論 Information Technology Essentials	3	選 (E)	一 Freshman	需先修計算機概論 prerequisite: Introduction to Computer Science
210156	JavaScript程式設計 JavaScript Programming	3	選 (E)	一 Freshman	需先修程式設計 prerequisite: Programming
210157	企業實習240(一) Enterprise Internship 240(一)	3	選 (E)	四 Senior	
210158	企業實習240(二) Enterprise Internship 240(二)	3	選 (E)	四 Senior	
210159	企業實習(二) Corporate Internship (2)	6	選 (E)	四 Senior	
210161	Linux系統管理 Linux System Administratio	3	選 (E)	一 Freshman	

210163	電腦與資訊安全概論 Introduction to Computer and Information Security	3	選 (E)	— Freshman	
210164	音樂,大腦,與科技 Music,Brain,Science and Technology	2	選 (E)	— Freshman	
210165	英文寫作, 口說, 簡報 (一) On English Writing, Speaking and Presentation (1)	3	選 (E)	— Freshman	
210166	數位音樂學習與創作 Digital Music Learning and Creation	2	選 (E)	— Freshman	
210167	Python 網路程式設計 Python Network Programming	3	選 (E)	二 Sophomore	
210168	程式競賽練習 Practice of Competitive Programming	3	選 (E)	— Freshman	
210169	C++程式設計實戰 C++ Programming Practice	3	選 (E)	— Freshman	
210170	弗列斯科網頁程式設計 Flask: Web Development with Python	3	選 (E)	— Freshman	
210171	程式設計與美學創作 (一) Creative Coding and Aesthetic Programming (1)	3	選 (E)	— Freshman	
210172	資訊科學特論 Special Topics in Computer Science	3	選 (E)	— Freshman	
備註: Notes: 專業選修42學分，3年級以上可選修系上碩士班課程作為專業選修。					